US Car Accident Analysis

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1. Introduction

The problem we aimed to address in our project was if it was possible determine both the risk of driving in different environments such as location, time, weather, and population. Another goal was to determine if an accident were to occur can we reliably predict how serious the resulting accident was on a scale of 1 (dent) to 4 (life threatening pileups). We found this problem interesting, as it could help determine the risk for off campus students driving to class as well as general driving risk. This information could further be used to aid in predicting routes that have the largest likely hood to have a severe pile up accident thus, this information could be used for route planning.

2. About The Data

Our data came from a Kaggle data set titled **US Accidents (2016-2023)** and is authored by **Sobhan Moosavi**. You can access the data set with the following link (Link to Dataset). The data was collected in 49 US states using methods such as using API's from US and state departments of transportation, law enforcement agencies, traffic cameras, and traffic sensors within road networks

3. Features Used

Features from the Database

- Severity: This is the accident severity label, ranging from 1 to 4, where higher values indicate more severe accidents.
- Start_Time: This is the date and time when the accident began.
- End_Time: This is the date and time when the accident ended.
- Start_Lat: This is the latitude of the accident location.
- Start_Lng: This is the longitude of the accident location.
- Street: This is the street name associated with the accident.
- City: This is the city where the accident occurred.
- County: This is the county where the accident occurred.
- State: This is the U.S. state abbreviation for the accident location.
- Zipcode: This is the ZIP code of the accident location.
- Country: This is the country code of the accident location.
- Temperature(F): This is the air temperature in degrees Fahrenheit at the time of the accident.
- Wind_Chill(F): This is the perceived temperature in degrees Fahrenheit after accounting for wind.
- Humidity(%): This is the humidity percentage at the time of the accident.
- Pressure(in): This is the air pressure in inches.
- Visibility(mi): This is the visibility in miles.
- Wind_Speed(mph): This is the speed of the wind in miles per hour.
- Precipitation(in): This is the amount of precipitation in inches.

Engineered Features

- hour: This is the hour of the day extracted from the accident start time.
- day_of_week: This is the day of the week extracted from the accident start time.
- month: This is the month of the year extracted from the accident start time.
- is_weekend: This indicates whether the accident occurred on a weekend.
- is_rush_hour: This indicates whether the accident occurred during a defined rush-hour period.
- lat_bin: This is a binned version of latitude used to group nearby accident locations.
- lng_bin: This is a binned version of longitude used to group nearby accident locations.
- bad_weather: This indicates whether weather conditions were poor based on visibility, precipitation, or wind speed.
- freezing: This indicates whether the temperature was below freezing.

4. Methods and Results

Accident Severity Model

For the accident severity prediction, we used a Random Forest classification. We selected this model because our data set was large and contained both categorical and continuous data. Random forests are able to deal with large data sets that contain both these types of data since they use trees to determine the classification of data. Each tree has branches nodes that split into different branches. Each branch in each split represents whether based on the current node (which represents that feature of the current sample) what category was that sample labeled as. So each tree keeps track of counts for classes based on the feature values in the tree. Therefore, throughout the tree we can have a mixture of categorical values and continuous values. The model uses many different versions of trees (e.g a random forest of trees) to prevent over fitting, this also helps with noise reduction. Random Forests was also a good choice because they are good at finding non linear relationships within the data. This is helpful for our project, as we do not expect a linear relationship between the severity of an accident and precipitation. For example, a light rain might slightly increase severity, while a heavy rain might exponentially increase severity. We handled missing data to improve the models stability. To deal with missing data we did the following, we handled missing values such as weather by using median imputation and missing-indicator flags.

We evaluated the final Random Forest classifier on the test set and the overall performance metrics were as follows:

- Accuracy: **0.76**
- Weighted Precision: **0.80**
- Weighted Recall: **0.76**
- Weighted F1-score: **0.77**

These results indicate solid overall performance, although class-level results show weaker performance on minority severity classes. We noticed that the F-1 score for accident severity of rating of 2 was 0.84 while severity of 1 and 4 had a score of 0.31 and 0.37 respectively. A large proportion of accidents were labeled as severity 2 (1,086,927 accidents) and only a small proportion were labeled as 1 (13,386 accidents) or 4(36,975 accidents). This leads us to believe that due to the imbalance in counts for severity classes the model is good at predicting severity 2 cases while the model struggles with 1 and 4 due to the lack of examples to train on. Our conclusion is that the imbalance between the number of severity classes is one of the limiting factors in the accuracy of the model but with more balanced data we believe that the model would find better results and thus Random Forest classification could be a good model to use for this problem in the future.

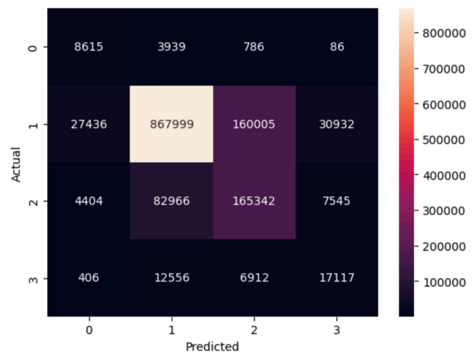

Figure 1. Confusion matrix for the accident severity classification model with x and y-axis representing accident severity rankings.

We also ran feature importance and found the following.

- Location-based features (`Start_Lng`, `Start_Lat`, `lng_bin`, `lat_bin`) were the most important predictors in the model.
- Seasonal and time-based features such as `month`, `hour`, and `day_of_week` also played a large role.
- Weather features including `Wind_Chill(F)`, `Pressure(in)`, `Temperature(F)`, and `Humidity(%)` had a moderate effect.
- Missing-value indicators also showed some importance, which suggests that missing weather data may itself be related to accident risk.
- `Visibility(mi)` had the smallest impact out of the top features.

Accident Risk Modeling with U.S. Census Data

For this section of the project, we built a model that can estimate the risk of driving based on location, time, and population data. For this we used real U.S. Census data for ZIP code locations and also included the population for each ZIP code. The main features used for this model were latitude, longitude, hour, day of week, month, and log-transformed population.

One limitation we have in this model is that the dataset we used only had real crash data and did not include non-crash data. Due to this, we created synthetic non-crash data. There was a dataset that had real world data for non crashes, but it would have cost us money to use that's why we generated our own data. We generated 150,000 synthetic non-crash data points and combined them with 150,000 real crash data points. We then trained a HistGradientBoostingClassifier on the data. We chose this model because it works well with the same types of data as random trees as mentioned above while also being less memory-intensive than Random Forest. Memory use was an important factor for us to consider since we ran into memory limits in Google Colab when trying Random Forest models.

One important issue to consider is that the dataset we created does not fully reflect the real world, since crashes are much rarer than non-crash events. This means the raw probabilities from the model would be too high if we used them directly since our data set contains a 50/50 split of crashes to non crashes. To fix this, we calibrated the model's predicted probabilities and then applied a prior adjustment using a small assumed baseline. This helped make the final outputs more realistic as risk estimates. Since the data is synthetic, it should only be used as an estimate for risk and not true risk since our model does not use a large amount of real world data.

We evaluated the final model on the test set using ROC-AUC, PR-AUC, and Brier score. The results were as follows:

- ROC-AUC: **0.7508**
- PR-AUC: **0.7379**
- Brier score: **0.2024**

Therefore, we can see that the ROC-AUC score tells us that our model is decent at differentiating if an input was an accident or not an accident. The PR-AUC score tells us that our model is doing a decent job of not only finding positive or negative cases but is actually moderately precise when doing so. The Brier score tells us that the model doesn't give the most accurate probabilities.

Even though we needed to use synthetic data we think this section of the project was useful in showing that HistGradientBoostingClassifier can be useful in predicting risk of driving and it might be useful to look deeper into if eventually we are able to get real world data.

5. Conclusion

After completing this project we learned that random forests could be useful in classifying accident severity if we can find more balanced data. We saw that our model accurately predicted level 2 severity accidents. Therefore, with enough information about a particular class the model was able to classify it accurately. The reason the model was not as good at predicting level 1 and 4 is because there was a lack of data for these cases. For the accident risk model we achieved reasonable results but not good enough that we would trust the model to make real world risk assessments especially since some of the data was synthetic. However, with real world non-crash data we think that our model would have preformed even better and thus could then be used to make real world driving risk assessments.

Acknowledgements

AI was consulted as a resource and not used to generate any part of the results. While snippets of code may have been adjusted and our project plan did need to pivot, the authors take full responsibility for our results, code, and project. The model created for this project is clearly limited in its applications and by no means a guarantee of safety or tragedy when driving in the United States. Use of AI throughout the project:

- Google. (2026). Gemini (Apr 20 version) [Large language model]. https://gemini.google.com/
  - Used in debugging Colab environment, syntax errors.
- OpenAI. (2023). ChatGPT (Mar 14 version) [Large language model]. https://chat.openai.com/chat
  - Consulted for project organization, filled in technical gaps.
- Anthropic. (2024). Claude 3.5 Sonnet [Large language model]. https://claude.ai
  - Code consultation.

Works Cited

- Moosavi, Sobhan. "US Accidents (2016–2021)." Kaggle. https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents